Module/Pathway Review: L-Valine, L-Leucine and L-Isoleucine Biosynthesis (KEGG ppu00290) in *Pseudomonas putida* KT2440

Taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Target bucket:** KEGG ppu00290 "Valine, leucine and isoleucine biosynthesis" **Scope of evidence:** UniProtKB (2026), KEGG (2026), and PubMed literature. All KT2440 ilv/leu proteins are UniProt evidence level PE=3 ("inferred from homology"); no target-strain enzymology exists for individual proteins. The strongest direct KT2440 evidence is a genome-wide auxotrophy screen (P 20158506).

---

1. Executive summary

The BCAA-biosynthesis module is **fully satisfiable** in *P. putida* KT2440: every one of the six canonical mechanistic steps has at least one strong, orthology-supported candidate, and de-novo amino-acid biosynthesis is experimentally required for growth on glucose minimal medium in this strain (P 20158506).

The catalytic trunk (`ilvIH` → `ilvC` → `ilvD`), the leucine extension branch (`leuA` → `leuCD` → `leuB`), and the terminal transaminase (`ilvE`) are unambiguous. The only genuinely ambiguous step is the isoleucine-entry threonine deamination, which is served by **two redundant biosynthetic paralogs** (`ilvA-I`/PP_3446 and `ilvA-II`/PP_5149).

The candidate list, however, needs curation. It (a) appears to "omit **PP_3365**," but this is a locus-level artifact — PP_3365 and the listed **PP_1157** encode an **identical** AHAS large-subunit protein (single UniProt entry **Q877U6**, two ordered-locus names), so no distinct enzyme is missing; (b) contains **four over-propagated members** that are not part of BCAA biosynthesis

— `alaA` (alanine), `ldh` (glutamate-dehydrogenase-like / BCAA *degradation*), and the short ACT-less dehydratases PP_3191 and PP_4430 (catabolic threonine/serine dehydratases), plus `PP_2930`/sdaA (serine ammonia-lyase); and (c) carries **incorrect primary-bucket labels** for `leuBCD` and `ilvHI` (tagged ppu00660, a KEGG map-overlap artifact — they are canonical ppu00290 genes). No lineage-specific citramalate (`cimA`) bypass exists, so threonine deamination is the sole isoleucine entry.

---

2. Target-organism pathway definition

Included biochemistry (ppu00290): conversion of pyruvate (and, for isoleucine, threonine-derived 2-oxobutanoate) to the three proteinogenic branched-chain amino acids via: 1. L-threonine → 2-oxobutanoate (threonine deaminase; isoleucine branch only) 2. 2× pyruvate → (S)-2-acetolactate, and pyruvate + 2-oxobutanoate → (S)-2-aceto-2-hydroxybutanoate (AHAS) 3. acetohydroxy acids → 2,3-dihydroxy acids (ketol-acid reductoisomerase) 4. 2,3-dihydroxy acids → branched 2-oxoacids (dihydroxy-acid dehydratase) 5. leucine-only extension: 2-oxoisovalerate → 2-isopropylmalate → 3-isopropylmalate → 2-oxoisocaproate (LeuA/LeuCD/LeuB) 6. transamination of all three 2-oxoacids → L-Val/L-Leu/L-Ile (BCAT, IlvE)

Neighboring maps to keep separate: - **ppu00280** "Valine, leucine and isoleucine *degradation*" — catabolism; shares `ilvE` (reversible transaminase) and contains `ldh` (PP_4617). - **ppu00660** "C5-branched dibasic acid metabolism" and **ppu00650** "Butanoate metabolism" — overlap on the *reaction* level (acetolactate, isopropylmalate), which is why `ilvHI`, `leuBCD`, and the catabolic acetolactate synthases co-map there. Map overlap ≠ separate biological function. - **ppu00260** "Glycine, serine and threonine metabolism" — where KEGG also lists all four K01754 threonine dehydratases and `sdaA`. - **ppu00250** / **ppu00270** (Ala/Asp/Glu; Cys/Met) — source of the `alaA` and `sdaA` over-annotations.

Alternate names / DB definitions: KEGG modules **M00019** (Val/Ile biosynthesis, pyruvate ⇒ valine / 2-oxobutanoate ⇒ isoleucine) and **M00432** (leucine biosynthesis) / **M00570** (isoleucine biosynthesis, threonine ⇒ 2-oxobutanoate ⇒ isoleucine). UniProt uses per-product "L-valine / L-leucine / L-isoleucine biosynthesis" pathway strings. BioCyc/MetaCyc splits the same chemistry into ILEUSYN-PWY, VALSYN-PWY, LEUSYN-PWY.

---

3. Expected step model and satisfiability calls

#	Step (enzyme / EC)	KT2440 gene(s)	Call
1	L-Thr → 2-oxobutanoate — threonine deaminase (4.3.1.19)	**ilvA-I PP_3446**, **ilvA-II PP_5149**	**covered** (redundant paralogs)
2	AHAS (2.2.1.6)	**ilvI PP_4680** + **ilvH PP_4679**	**covered**
3	Ketol-acid reductoisomerase (1.1.1.86)	**ilvC PP_4678**	**covered**
4	Dihydroxy-acid dehydratase (4.2.1.9)	**ilvD PP_5128**	**covered**
5a	2-isopropylmalate synthase (2.3.3.13)	**leuA PP_1025**	**covered**
5b	Isopropylmalate isomerase (4.2.1.33)	**leuC PP_1985** + **leuD PP_1986**	**covered**
5c	3-isopropylmalate dehydrogenase (1.1.1.85)	**leuB PP_1988**	**covered**
6	Branched-chain aminotransferase (2.6.1.42)	**ilvE PP_3511**	**covered**

All six steps covered. No gaps and no `not_expected_in_target_taxon` steps: KT2440 is a BCAA prototroph.

---

4. Candidate genes and evidence

High-confidence biosynthetic members (promote/keep as covered): - ilvI (PP_4680, Q88DY8) + **ilvH (PP_4679, Q88DY9)** — anabolic AHAS; UniRule pathway "L-valine/L-isoleucine biosynthesis step 1"; ilvH carries the ACT valine-feedback domain. Organised in the **ilvIH-ilvC operon (PP_4678-4680)** — the strongest operon evidence in this module. - **ilvC (PP_4678, Q88DZ0)** — type-I KARI, EC 1.1.1.86; detailed catalytic FUNCTION annotation. - **ilvD (PP_5128, Q88CQ2)** — dihydroxy-acid dehydratase, EC 4.2.1.9. - **leuA (PP_1025, Q88P28)** — 2-isopropylmalate synthase; UniProt "L-leucine biosynthesis step 1/4". Genomic orphan. - **leuC (PP_1985)/leuD (PP_1986)** — isopropylmalate isomerase large/small subunits, clustered. - **leuB (PP_1988, Q88LE5)** — 3-isopropylmalate dehydrogenase. - **ilvE (PP_3511, Q88H54)** — BCAT;

UniProt assigns it to all three L-Val/L-Leu/L-Ile pathways; also appears in the ppu00280 degradation map because the transamination is reversible (expected, not an error). Single-copy — no paralog backup, so functionally pivotal. - **ilvA-I (PP_3446, Q88HB4)** and **ilvA-II (PP_5149, Q88CN1)** — both ~500–530 aa with two C-terminal **ACT-like regulatory domains**, UniProt-named "L-threonine dehydratase" with the isoleucine-biosynthesis pathway. Both are genomic orphans. Redundant biosynthetic paralogs.

Copy-number check (KEGG ortholog counts in KT2440): the trunk/branch enzymes are all single-copy — ilvE/BCAT (K00826, PP_3511), leuA (K01649, PP_1025), leuB (K00052, PP_1988), leuCD (K01703, PP_1985), ilvD (K01687, PP_5128), ilvC (K00053, PP_4678) — so there is no paralog ambiguity at these steps; ilvE in particular has no backup and is functionally pivotal. Critically, the **AHAS small (regulatory) subunit is also single-copy** (K01653, ilvH/PP_4679): since the other three K01652 large subunits (Q877U6=PP_1157/PP_3365; PP_1394) have no cognate small subunit, only ilvI can assemble the anabolic, valine-feedback-regulated AHAS holoenzyme — independent support that ilvIH is the sole biosynthetic AHAS and the other large subunits are catabolic/independent acetolactate synthases.

Evidence type for all of the above: homology / UniRule + KEGG orthology + (for ilvIH-ilvC, leuCD) operon context. Direct enzymology is *not* available for any KT2440 protein; whole-module function is supported in vivo by auxotrophy on minimal medium (P 20158506).

---

5. Gaps, ambiguities, and likely over-annotations

Paralog ambiguity (candidate_uncertain, needs experimental disambiguation): - ilvA-I vs ilvA-II — which paralog carries the physiological biosynthetic load (and whether one is isoleucine-feedback-regulated while the other is not) is unresolved. Redundancy explains why neither would necessarily show an auxotrophic phenotype when singly disrupted. - **AHAS large-subunit paralogs** — besides anabolic `ilvI`, KT2440 encodes additional K01652 large subunits. Crucially, **PP_1157 and PP_3365 are two loci encoding an identical 547-aa protein** (single UniProt entry **Q877U6**; confirmed by KEGG DBLINKS and UniProt `orderedLocusNames`), so the apparent "missing PP_3365" is a duplicate gene copy, not a new enzyme. The distinct paralogs are therefore Q877U6 (PP_1157/PP_3365) and **PP_1394 (Q88N22)**. Neither has a UniProt BCAA-pathway assignment or an adjacent small regulatory subunit, and both map primarily to ppu00660/00650. Sequence classification (3-mer cosine vs characterized references) shows **ilvI/PP_4680 is unambiguously the anabolic AHAS** (top match to *E. coli* anabolic IlvI, 0.43, vs ≤0.26 for all others), while Q877U6 and PP_1394 are divergent from all anabolic references (0.16–0.22) → most likely **catabolic acetolactate synthases** (acetoin/2,3-butanediol), candidate_uncertain, leaning not-in-biosynthesis-module.

Likely over-propagation (should NOT be biosynthetic-module members): - **PP_3191 (Q88I11, 350 aa)** and **PP_4430 (Q88EM4, 330 aa)** — ACT-less, UniProt-unnamed PALP-fold proteins auto-mapped to K01754/EC 4.3.1.19. The ~330 aa length matches catabolic threonine dehydratase (E. coli TdcB ≈328 aa); genomic neighbours are amino-acid-catabolism genes (ornithine cyclodeaminase, GntR/D-Arg regulators). Broad EC 4.3.1.19 mapping, not biosynthetic evidence. - **PP_2930 (Q88IR9)** — L-serine ammonia-lyase (sdaA, EC 4.3.1.17); serine, not threonine/BCAA. Over-annotation into ppu00290. - **alaA (PP_1872, Q88LQ7, EC 2.6.1.2)** — glutamate-pyruvate aminotransferase, UniProt pathway "L-alanine biosynthesis". Not BCAA. Over-propagation (transaminase EC overlap). - **ldh (PP_4617, Q88E51, EC 1.4.1.9)** — annotated "leucine dehydrogenase," but UniProt domain is Glu/Phe/Leu/Val/Trp-dehydrogenase and FUNCTION is glutamate oxidative deamination; KEGG places it in ppu00280 **degradation**. Even if it can reductively aminate 2-oxoisocaproate, the canonical biosynthetic terminal step is IlvE transamination — treat as catabolic/alternative, not a core biosynthetic member.

Metadata bucket errors (module_needs_revision): - `leuC/leuD/leuB` (PP_1985/86/88) and `ilvH/ilvI` (PP_4679/80) are labeled *primary bucket kegg:ppu00660*. They are canonical ppu00290 biosynthetic genes; the 00660 tag is a KEGG map-overlap artifact and should be corrected to primary ppu00290.

---

6. Module and GO-curation recommendations

- **Mark covered:** steps 1–6 (see §3). The module is satisfiable; KT2440 is a BCAA prototroph.
- **candidate_uncertain:** `ilvA-I`/`ilvA-II` (which is dominant); `PP_3365`, `PP_1157`, `PP_1394` (AHAS-family paralogs of uncertain anabolic role); `ldh` (catabolic vs alternative aminating route).
- **Remove from biosynthesis module (over-annotation):** `alaA` (PP_1872), `PP_2930` (sdaA), and the short catabolic dehydratases `PP_3191`, `PP_4430`. `ldh` (PP_4617) belongs primarily to ppu00280.
- **module_needs_revision:**

- Document **PP_3365** as a duplicate locus of PP_1157 (identical protein, UniProt Q877U6) — this reconciles the KEGG 19-gene vs metadata 18-gene count without adding a new enzyme. Both loci should be marked candidate_uncertain (catabolic-ALS-leaning), consistently bucketed.
- Re-bucket `leuBCD` and `ilvHI` to primary **ppu00290**.
- Record that `ilvE` co-membership in ppu00280 is expected (reversible transaminase), not an error.
- **Module boundaries:** the generic outline is essentially correct for KT2440. One refinement — the outline's "any validated member of the biosynthetic threonine-deaminase family satisfies the entry" should specify **ACT-domain-containing, full-length** (~500 aa) deaminases, to exclude the short catabolic K01754 hits.
- **GO curation:** no new GO terms required; existing terms suffice — GO:0009097 (isoleucine biosynthesis), GO:0009099 (valine biosynthesis), GO:0009098 (leucine biosynthesis), GO:0003984 (AHAS), GO:0004160 (DHAD), GO:0004084 (BCAT), GO:0004794 (L-threonine ammonia-lyase). Recommend GO annotations be restricted to the covered members above and *withheld* from the over-propagated candidates.

---

7. Genes to promote to full `fetch-gene` review

1. **ilvA-I (PP_3446)** and **ilvA-II (PP_5149)** — resolve which is the principal biosynthetic threonine deaminase and its feedback regulation (paralog redundancy).
2. **Q877U6 (PP_1157 = PP_3365 duplicate loci)** and **PP_1394** — determine anabolic vs catabolic (acetoin/butanediol) AHAS role; confirm the two loci are truly identical copies.
3. **ldh (PP_4617)** — confirm glutamate-dehydrogenase-like vs true leucine-dehydrogenase function and biosynthetic relevance.
4. **PP_1157 / PP_1394** — classify as catabolic acetolactate synthases (acetoin/butanediol) vs redundant AHAS.
5. (Lower priority) **PP_3191 / PP_4430** — confirm catabolic threonine/serine dehydratase assignment to justify removal from the biosynthesis module.

---

8. Evidence summary and open questions

- **Direct (target strain):** genome-wide minimal-medium auxotrophy screen shows amino-acid biosynthesis genes are conditionally essential in KT2440 (Molina-Henares et al. 2010, P 20158506) — establishes the module is functional and required in vivo, but does not resolve individual paralogs.
- **Homology / database inference:** all per-gene step assignments (UniProt UniRule PE=3, KEGG KO). Strong for the single-copy trunk/branch enzymes; weaker where multiple paralogs share a KO (K01754 deaminases; K01652 AHAS large subunits).
- **Species transfer:** the biosynthetic-vs-catabolic threonine-deaminase distinction (ACT-domain, length) is a robust, sequence-intrinsic criterion (E. coli IlvA vs TdcB) that transfers strongly. AHAS-isozyme physiology in other *Pseudomonas* (e.g., *P. aeruginosa* ilvBN/ilvIH) is a weaker transfer and should not be assumed identical in KT2440.
- **Open questions / resolving experiments:** (i) single vs double `ilvA-I/ilvA-II` knockouts on minimal medium ± isoleucine to test redundancy; (ii) does `PP_3365`/`ilvI` deletion cause valine/isoleucine auxotrophy (anabolic AHAS test)?; (iii) enzymatic assay or RB-TnSeq fitness (cf. P 38323821) to confirm PP_1157/PP_1394 are acetoin-pathway ALS; (iv) verify `ldh` (PP_4617) directionality in vivo.

Key references

- Molina-Henares AJ, et al. (2010) *Identification of conditionally essential genes for growth of Pseudomonas putida KT2440 on minimal medium…* Environ Microbiol. P 20158506. — direct KT2440 auxotrophy evidence.
- Borchert AJ, et al. (2024) *Machine learning analysis of RB-TnSeq fitness data predicts functional gene modules in [P. putida].*

P 38323821 .

— functional-genomics resource to disambiguate paralogs.

- Borchert AJ & Downs DM (2018) *Analyses of variants of the Ser/Thr dehydratase IlvA…2-aminoacrylate metabolism.*

P 30327426 .

— biosynthetic IlvA (EC 4.3.1.19) mechanism/context.

- UniProtKB entries: Q88DY8 (ilvI), Q88DY9 (ilvH), Q88DZ0 (ilvC), Q88CQ2 (ilvD), Q88P28 (leuA), Q88LE8/Q88LE7/Q88LE5 (leuC/D/B), Q88H54 (ilvE), Q88HB4/Q88CN1 (ilvA-I/ilvA-II).
- KEGG pathways ppu00290 / ppu00280 / ppu00660 / ppu00260; modules M00019, M00432, M00570.

Caveat: no per-protein enzymology exists for KT2440 BCAA-biosynthesis enzymes; all gene-level calls are homology/orthology-based, with whole-module function corroborated in vivo by minimal-medium auxotrophy.
